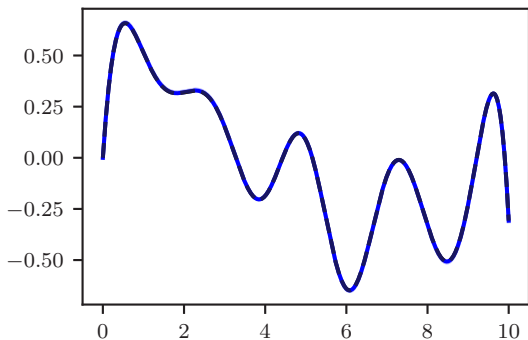


Second-Kind CM Trajectory Inversion $\tilde{y}(t) \mapsto x(t)$

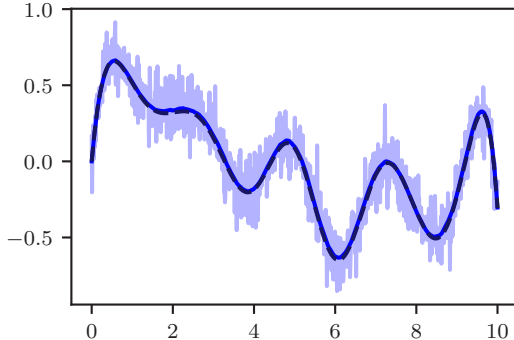
Reconstruct $x(t)$ from CM Volterra Eq.
with $c_0 = -2, c_1 = 0$ (noise = 0.00%)



$$\frac{\|\hat{x}_{\text{spec}}(t) - x(t)\|_{L^2}}{\|x(t)\|_{L^2}} = 2.843\text{e-}05$$

$$\frac{\|\hat{x}_{\text{data}}(t) - x(t)\|_{L^2}}{\|x(t)\|_{L^2}} = 9.802\text{e-}03$$

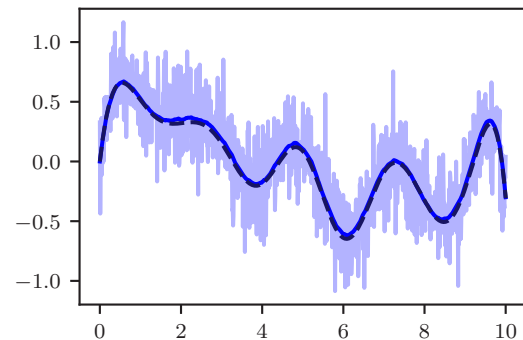
Reconstruct $x(t)$ from CM Volterra Eq.
with $c_0 = -2, c_1 = 0$ (noise = 50.00%)



$$\frac{\|\hat{x}_{\text{spec}}(t) - x(t)\|_{L^2}}{\|x(t)\|_{L^2}} = 3.472\text{e-}02$$

$$\frac{\|\hat{x}_{\text{data}}(t) - x(t)\|_{L^2}}{\|x(t)\|_{L^2}} = 3.270\text{e-}01$$

Reconstruct $x(t)$ from CM Volterra Eq.
with $c_0 = -2, c_1 = 0$ (noise = 100.00%)

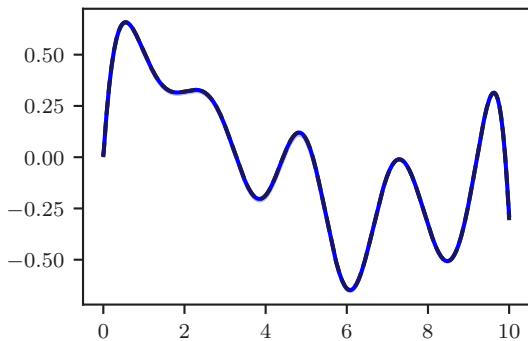


$$\frac{\|\hat{x}_{\text{spec}}(t) - x(t)\|_{L^2}}{\|x(t)\|_{L^2}} = 6.944\text{e-}02$$

$$\frac{\|\hat{x}_{\text{data}}(t) - x(t)\|_{L^2}}{\|x(t)\|_{L^2}} = 6.535\text{e-}01$$

First-Kind PD Trajectory Inversion $\tilde{y}(t) \mapsto x(t)$

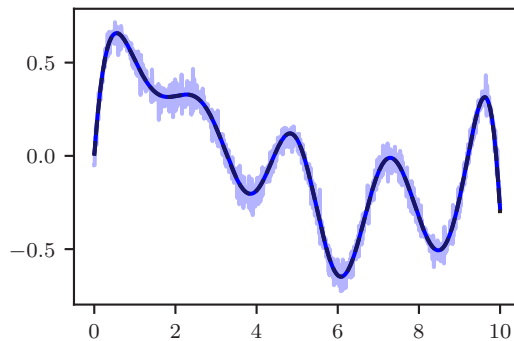
Reconstruct $x(t)$ from PD Volterra Eq.
with $c_0 = 0, c_1 = 0$ (noise = 0.00%)



$$\frac{\|\hat{x}_{\text{spec}}(t) - x(t)\|_{L^2}}{\|x(t)\|_{L^2}} = 2.590\text{e-}03$$

$$\frac{\|\hat{x}_{\text{data}}(t) - x(t)\|_{L^2}}{\|x(t)\|_{L^2}} = 1.796\text{e-}02$$

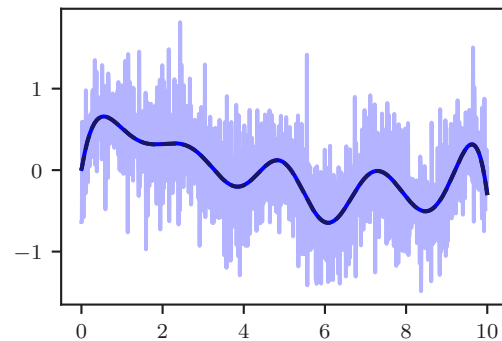
Reconstruct $x(t)$ from PD Volterra Eq.
with $c_0 = 0, c_1 = 0$ (noise = 0.10%)



$$\frac{\|\hat{x}_{\text{spec}}(t) - x(t)\|_{L^2}}{\|x(t)\|_{L^2}} = 2.637\text{e-}03$$

$$\frac{\|\hat{x}_{\text{data}}(t) - x(t)\|_{L^2}}{\|x(t)\|_{L^2}} = 1.305\text{e-}01$$

Reconstruct $x(t)$ from PD Volterra Eq.
with $c_0 = 0, c_1 = 0$ (noise = 1.00%)



$$\frac{\|\hat{x}_{\text{spec}}(t) - x(t)\|_{L^2}}{\|x(t)\|_{L^2}} = 5.397\text{e-}03$$

$$\frac{\|\hat{x}_{\text{data}}(t) - x(t)\|_{L^2}}{\|x(t)\|_{L^2}} = 1.293\text{e+}00$$

--- $x(t)$
— $\hat{x}_{\text{spec}}(t)$
— $\hat{x}_{\text{data}}(t)$